

Intelligent Elephant Care And Visitor Insights: A Multi-Component Research Study at Pinnawala Elephant Orphanage

Project ID: 25-26J-391

Progress Presentation 2 (PP2)

2026.03.09



Introduction & Overview



At Pinnawala Elephant Orphanage

Problem Statement & Stakeholders

Context

Pinnawala Elephant Orphanage currently relies on manual observation and traditional record-keeping to monitor elephant health, stress, feeding, and visitor activities. These methods can lead to delayed detection of health issues, inaccurate stress assessment, and inefficient management decisions.

Stakeholders

PRIMARY

Elephant Caretakers
Veterinary Doctors
Orphanage Management

SECONDARY

Visitors / Tourists



Observed Issues



Manual Monitoring

Elephant health, stress levels, and feeding decisions are based on manual observation by caretakers, which can be subjective and inconsistent.



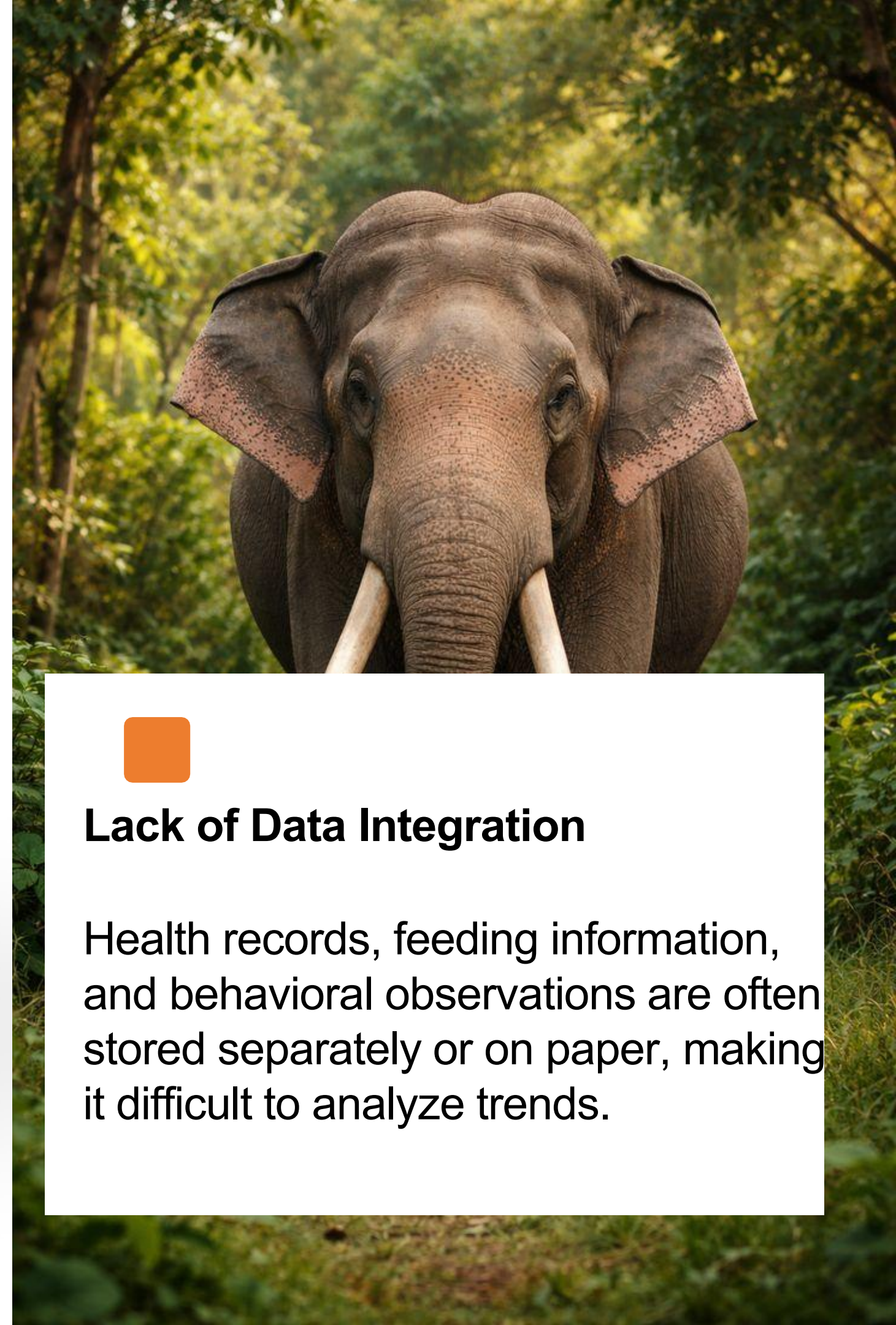
Delayed Detection of Problems

Early signs of illness or stress may not be detected quickly, leading to delayed treatment and reduced elephant welfare.



Lack of Data Integration

Health records, feeding information, and behavioral observations are often stored separately or on paper, making it difficult to analyze trends.





MAIN OBJECTIVE

To develop an intelligent and integrated system that improves elephant welfare and visitor management at Pinnawala Elephant Orphanage by using AI-based technologies for health monitoring, stress detection, personalized feeding recommendations, and visitor behavior analytics.

Research Gap & Objectives

RESEARCH GAP

Current elephant management systems mainly rely on manual observation and separate data records, which do not provide real-time monitoring or predictive insights. There is a lack of an integrated, intelligent system that combines health monitoring, stress detection, feeding recommendations, and visitor analytics to support better decision-making and improve elephant welfare.

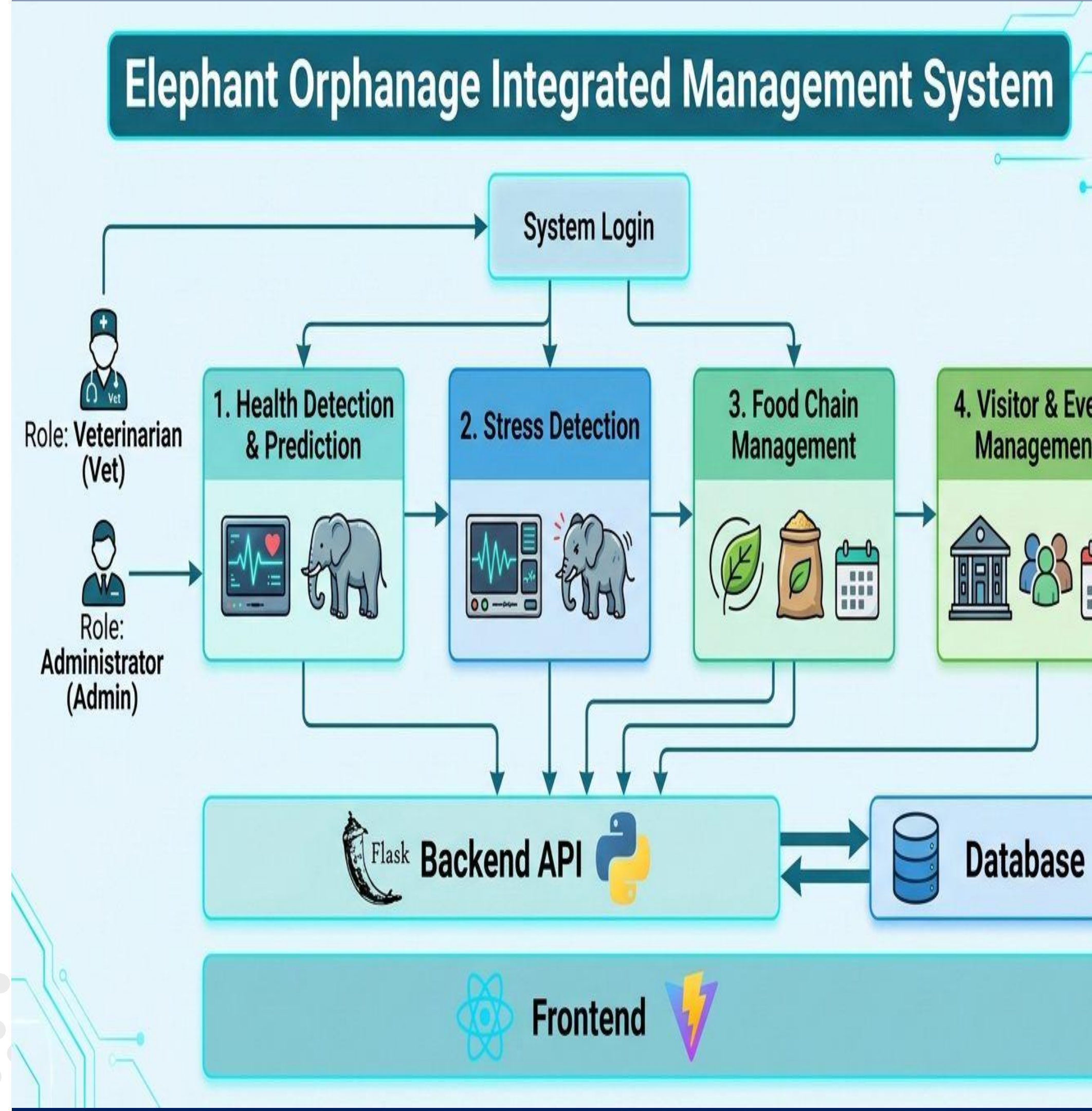


Solution Overview

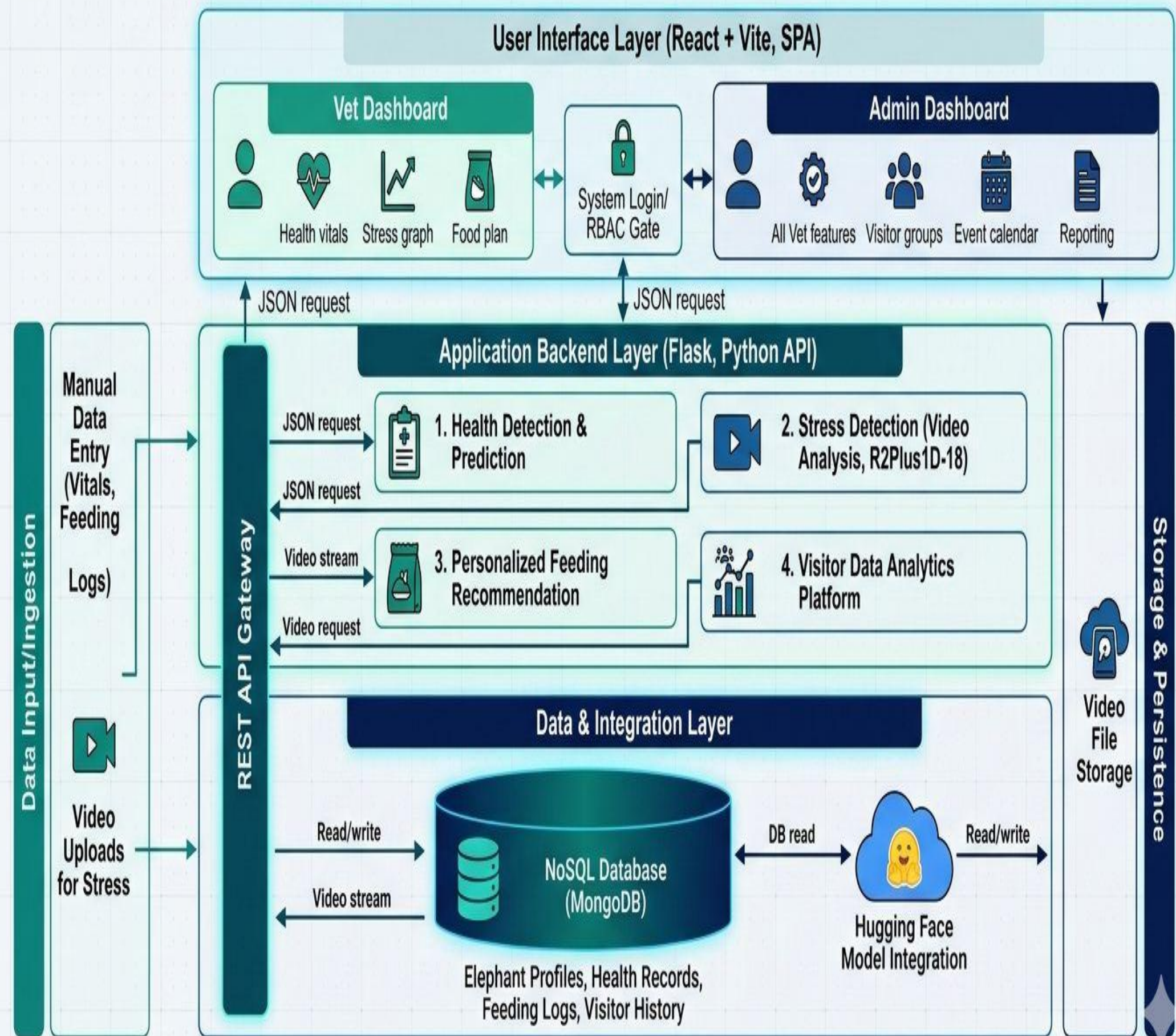
SUMMARY

Integrated platform: Elephant Health Detection & Prediction Platform , Elephant Stress Detection Platform, Personalized Elephant Feeding Recommendation Platform, and a visitor Data Analytics Platform.

- AI-powered dual-model pipeline analyzing twelve vitals for life-saving detection.
- Elephant Stress Detection using R2Plus1D-18 Video Action Recognition Model
- Intelligent system recommending optimal elephant diets using health and activity data.
- The collected data is analyzed and shown on a **management dashboard** to identify popular events and optimize visitor scheduling.



Elephant Care AI Management System Architecture



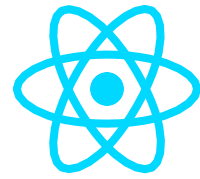
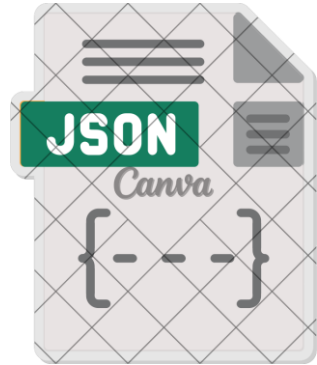
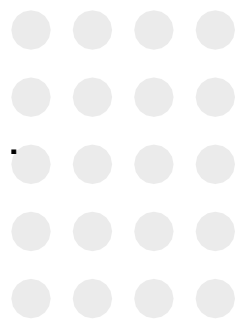
System Architecture

Component A (C1):
FERNANDO K.A.S.E
IT22591920

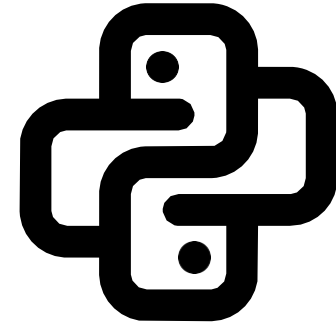
Component B (C2):
NARASINGLHE P.A.I.D
IT22590480

Component C (C3):
WIJERATHNE K.W.W.M.A.K
IT22031402

Component D (C4):
SENADHEERA S.P.K.N
IT22316004



React



Integration Points

API S / DATA FORMATS

APIs: Using Flask python and React vite for communication between system component.

Database: SQLite (Serverless, local file-based database).
MongoDB for store visitor data.

Data Format: JSON is used for structure data exchange between all components.

MODEL ACCESS

M1:- elephant_health_model.pkl – Accessed using predict() from core_logic.health_predictor

diagnosis_model.pkl – Accessed through the predict() function in core_logic.health_predictor to determine the elephant's possible illness from health data.

M2:-Model: R2Plus1D-18 (PyTorch) — Deployed via Hugging Face Space for Video-Based Stress Detection

M3:- from core_logic.food_quantity_engine import recommend_food; result = recommend_food(elephant_data, daily_vitals)

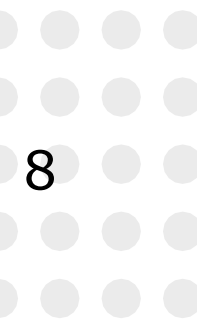
M4:- The system uses a Random Forest Regressor to predict visitor patterns. The results are shown on the dashboard to support management decisions



Hugging Face



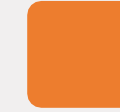
TensorFlow





PERFORMANCE

The system should process data (health, stress, feeding, visitor data) and generate insights in near real-time.



USABILITY

Interfaces should be simple for caretakers, veterinarians, and management staff to operate.



Non-Functional Targets & Status



SECURITY & PRIVACY

Elephant health records and visitor data should be securely stored and managed.



RELIABILITY

The system should continuously collect and analyze data without frequent failures.



Current Completion Status: ~90%+ across all four components

Commercialization / Impact



VALUE PROPOSITION

An integrated AI-based platform that improves elephant welfare through health prediction, stress detection, feeding optimization, and visitor analytics.

TARGET USERS/MARKET

Elephant orphanages,
wildlife conservation
authorities, zoos, national
parks, and tourism
management organizations.

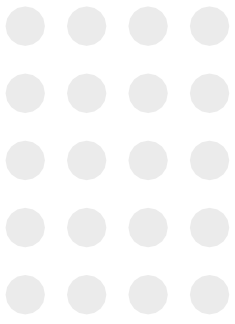


COST & FEASIBILITY

Cost-effective system using
cameras and cloud platforms;
deployable using existing
infrastructure with potential
subscription and institutional
funding models.

Adoption plan

Phase	Description	Purpose
Pilot Deployment (3 – 6 mon)	Test the integrated system at Pinnawala Elephant Orphanage using real data.	Evaluate system performance in a real environment.
System Integration	Improve AI models and integrate the system with existing orphanage operations.	Enhance accuracy, usability, and reliability.
Expansion	Deploy the system in zoos, wildlife parks, and conservation centers	Enable wider adoption and improve elephant welfare



**Thank
You !**

